

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a80add4737793b416.jsonl`  
- SHA-256: `4bdc0966d8a6bd6a93a65e3d8e033f5a856475e491787df5551f8cc874382bb4`  
- Source modified: `2026-05-31T04:25:59+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:24:30.213Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"The Surya CODE and the Surya WEIGHTS are licensed separately: the repo's code LICENSE file is Apache-2.0 (permissive), but the MODEL_LICENSE explicitly excludes source code/scripts ('Complementary Material') from the OpenRAIL-M terms. Verifying the repo license alone would wrongly conclude Surya is fully permissive ? the weights are the binding constraint, and the current weights license is OpenRAIL-M, NOT the GPL/commercial dual-license that older descriptions cite."

****Source:**** https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)
****Supporting quote:**** "'Complementary Material' means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model, and used to prepare data for training or evaluation, if any."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:24:32.762Z)

I'll verify this claim about Surya's licensing. Let me fetch the primary source and check the actual license files.

3. assistant (2026-05-31T04:24:37.055Z)

I'll fetch the primary source MODEL_LICENSE file and the repo's code LICENSE in parallel, plus search for contradicting evidence on Surya's current licensing.

4. user (2026-05-31T04:24:42.559Z)

License Analysis

The file contains the **Apache License, Version 2.0**.

The opening lines state:

```
"Apache License
Version 2.0, January 2004
http://www.apache.org/licenses/"
```

This is indeed Apache-2.0, not GPL or another license type.

5. user (2026-05-31T04:24:42.796Z)

License Analysis

License Name: AI PUBS OPEN RAIL-M LICENSE (MODIFIED)

This is a modified version of the OpenRAIL-M license, as indicated in the document's title and Section I preamble.

Definition of Complementary Material:

"Complementary Material means the applicable source code and scripts used to define, run, load, benchmark or evaluate the Model... This includes any accompanying documentation, tutorials, examples, and any related information, if any. Complementary Material is not licensed under this License."

Source Code/Scripts Exclusion:

The license explicitly states that source code and scripts are excluded from the model license terms. Complementary Material (which includes source code and scripts) operates under separate licensing arrangements, not under this OpenRAIL-M modification.

6. user (2026-05-31T04:24:47.445Z)

Web search results for query: "Surya OCR datalab-to license OpenRAIL-M model weights GPL commercial 2025"

Links: [{"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}, {"title": "surya/MODEL_LICENSE at master · datalab-to/surya", "url": "https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE"}, {"title": "surya/README.md at master · datalab-to/surya", "url": "https://github.com/datalab-to/surya/blob/master/README.md"}, {"title": "Surya: The Revolutionary OCR Toolkit for 90+ Languages", "url": "https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"}, {"title": "surya-ocr 0.17.1 on PyPI - Libraries.io - security & maintenance data for open source software", "url": "https://libraries.io/pypi/surya-ocr"}, {"title": "Datalab - Intelligent Document Processing: IDP-Software", "url": "https://idp-software.com/vendors/datalab/"}, {"title": "datalab-to/marker | zdoc.app", "url": "https://www.zdoc.app/en/datalab-to/marker"}, {"title": "pitapo/surya · Hugging Face", "url": "https://huggingface.co/pitapo/surya"}, {"title": "https://github.com/datalab-to/surya | Ecosystems: Awesome", "url": "https://awesome.ecosystems/projects/github.com/datalab-to/surya"}, {"title": "Overview - Datalab Documentation", "url": "https://documentation.datalab.to/docs/on-prem/overview"}]

Based on the search results, here's what I found about Surya OCR's licensing:

Licensing Details

Surya's model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$2M funding/revenue) and the code is GPL.

For broader commercial licensing or to remove GPL requirements, users can visit the pricing page.

The licensing structure is dual-tiered:

- 1. **Model Weights**: The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).
- 2. **Commercial Use**: Larger companies need a commercial license from Datalab to remove GPL code requirements and access enterprise support.

The project is maintained by Datalab (datalab-to) and was released in 2025 as "Surya: A lightweight document OCR and analysis toolkit" by Vikas Paruchuri and the Datalab Team.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-05-31T04:24:50.713Z)

The search results reveal a critical nuance. Let me examine the claim carefully against the evidence. The claim states the code LICENSE is Apache-2.0, but the search result says "the code is GPL." Let me read the actual repository README and license files directly to resolve this contradiction.

8. user (2026-05-31T04:24:55.715Z)

Licensing Information from Surya README

Here is the exact licensing section from the document:

Code License: "The Surya code is licensed under Apache 2.0."

Model Weights License: "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)."

Commercial Licensing: The document states: "For broader commercial licensing of the model weights, visit our pricing page [here](https://www.datalab.to/pricing?utm_source=gh-surya)."

The README makes a clear distinction: the source code uses Apache 2.0 (permissive open source), while the model weights use OpenRAIL-M with specific restrictions for commercial use. Organizations exceeding \$5M in funding or revenue must obtain commercial licensing separately.

9. user (2026-05-31T04:24:55.898Z)

License Analysis

First 15 Lines (Verbatim):

...
Apache License
Version 2.0, January 2004
<http://www.apache.org/licenses/>

TERMS AND CONDITIONS FOR USE, REPRODUCTION, AND DISTRIBUTION

1. Definitions.

"License" shall mean the terms and conditions for use, reproduction, and distribution as defined by Sections 1 through 9 of this document.

"Licensor" shall mean the copyright owner or entity authorized by the copyright owner that is granting the License.

"Legal Entity" shall mean the union of the acting entity and all other entities that control, are controlled by, or are under common control with that entity. For the purposes of this definition, "control" means

****License Type:****

This is ****Apache-2.0**** (Apache License Version 2.0).

****Exact License Header:****

"Apache License / Version 2.0, January 2004 / <http://www.apache.org/licenses/>"

10. assistant (2026-05-31T04:24:59.277Z)

The current repository README and LICENSE file directly confirm the claim's core assertions. Let me verify the historical "GPL or commercial" dual-license point and the older descriptions to confirm the claim's framing about what "older descriptions" cite.

11. user (2026-05-31T04:25:05.687Z)

License Information for Surya Repository

****Sidebar/About Section:**** The repository identifies itself as having an ****Apache-2.0 license****.

****README Code vs. Model Weights:****

The README explicitly distinguishes between two separate licenses:

1. ****Code License:**** "The Surya code is licensed under ****Apache 2.0****."
2. ****Model Weights:**** The weights use "a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)."

****GPL Mention:**** There is ****no mention of GPL**** anywhere in the provided content. The repository uses Apache 2.0 for code and a custom Open Rail-M variant for model weights?neither is GPL-based.

12. user (2026-05-31T04:25:06.644Z)

Web search results for query: "Surya OCR GPL-3.0 commercial license history "less than \$5M" datalab model weights change"

Links: [{"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}, {"title": "surya-ocr · PyPI", "url": "https://pypi.org/project/surya-ocr/"}, {"title": "surya/MODEL_LICENSE at master · datalab-to/surya", "url": "https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE"}, {"title": "datalab-to/surya-ocr-2 · Hugging Face", "url": "https://huggingface.co/datalab-to/surya-ocr-2"}, {"title": "surya/LICENSE at master · datalab-to/surya", "url": "https://github.com/datalab-to/surya/blob/master/LICENSE"}, {"title": "GitHub - CibiQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub", "url": "https://github.com/CibiQuadance/surya-OCR"}, {"title": "Releases · datalab-to/surya", "url": "https://github.com/datalab-to/surya/releases"}, {"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages", "url": "https://github.com/datalab-to/surya/tree/master"}, {"title": "surya-ocr 0.3.0", "url": "https://pypi.org/project/surya-ocr/0.3.0/"}, {"title": "surya/README.md at master · datalab-to/surya", "url": "https://github.com/datalab-to/surya/blob/master/README.md"}]

Based on the search results, here's what I found regarding Surya OCR's licensing and commercial license history:

Current Licensing Structure

The Surya code is licensed under Apache 2.0. However, there's an important distinction between the code and model weights: The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

License History Changes

The search results reveal that Surya's licensing terms have changed over time. An older version shows a different threshold: model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$2M funding/revenue) and our code is GPL. This indicates the revenue threshold was increased from \$2M to \$5M at some point.

Additionally, there's a reference to an earlier iteration of the project with different licensing: The weights are licensed cc-by-nc-sa-4.0, but I will waive that for any organization under \$5M USD in gross revenue in the most recent 12-month period.

Commercial Licensing

For broader commercial licensing of the model weights, visit our pricing page [here](#).

The project is maintained by Datalab (datalab-to), and the licensing terms clearly distinguish between free use for smaller organizations and commercial licensing options for larger enterprises.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. assistant (2026-05-31T04:25:22.145Z)

I now have enough evidence to evaluate the claim thoroughly. Let me assess each part against the verification checklist.

Evidence gathered:

- Code LICENSE = Apache-2.0:** Confirmed. The raw `LICENSE` file at `datalab-to/surya/master` begins verbatim with "Apache License / Version 2.0, January 2004". GitHub's sidebar also reports Apache-2.0. ? (claim correct)
- MODEL_LICENSE = OpenRAIL-M variant, excludes source/scripts:** Confirmed. The file is titled "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)". The "Complementary Material" definition matches the supporting quote verbatim, and the license explicitly states "Complementary Material is not licensed under this License" ? i.e., source code/scripts are excluded from the model-weights terms. ? (claim correct)
- Weights are OpenRAIL-M, NOT GPL-or-commercial dual-license:** Current README says weights "use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)." ? (claim correct that the *current* weights license is OpenRAIL-M)

Now the skeptical scrutiny ? where the claim becomes a problem:

The claim asserts a clean separation: "the repo's code LICENSE file is Apache-2.0 (permissive)... NOT the GPL/commercial dual-license that older descriptions cite." But my searches surfaced strong evidence that **the GPL was Surya's actual code license historically, not merely a misdescription**. An older README snapshot states verbatim: "our code is GPL" (with the \$2M threshold), and an even earlier version used **CC-BY-NC-SA-4.0** for weights. So:

- The claim frames "GPL/commercial dual-license" as something only "older descriptions cite" (implying it was always a misread). In reality, **GPL was genuinely Surya's code license in a prior era** ? the licensing materially changed over time (CC-BY-NC-SA ? GPL code + OpenRAIL-M ? Apache code + OpenRAIL-M). This is a real history, not a misdescription to be debunked.
- More importantly for the research question's purpose: the claim's takeaway is that "verifying the repo license alone would wrongly conclude Surya is fully permissive ? the weights are the

binding constraint." This core practical conclusion is **correct and well-supported**: code is Apache-2.0, but weights are OpenRAIL-M (commercial-use-restricted above \$5M, with field-of-use/AUP restrictions), so Surya's weights are NOT commercially-permissive (MIT/BSD/Apache) and must be flagged for muneem.

Let me confirm the OpenRAIL-M restrictions point (use restrictions / non-free-for-large-commercial) to be sure the "binding constraint" framing holds.

14. user (2026-05-31T04:25:31.220Z)

License Analysis: Surya MODEL_LICENSE

****Yes, this license contains use-based restrictions and commercial limitations.****

Commercial Use Restrictions

The license imposes revenue and funding thresholds that restrict commercial use:

1. ****Revenue threshold****: "for any purpose if You...generated more than two million US Dollars (\$5,000,000) in gross revenue"
2. ****Funding threshold****: "for any purpose if You...has raised more than two million US dollars (\$5,000,000) in total equity or debt funding"
3. ****Competitive restriction****: Organizations cannot use it "if You...provides or otherwise makes available any product or service that competes with any product or service offered by...Licensor"

These restrictions apply unless the use is "limited to personal use or research purposes."

Comparison to Permissive Licenses

This is **not** free for unrestricted commercial use like Apache-2.0 or MIT. Those licenses permit any commercial use without revenue or funding conditions. The Surya license's AI PUBS OPEN RAIL-M framework explicitly restricts larger organizations and competing products, requiring users to obtain commercial licenses from the licensor (datalab.to) for broader commercial deployment.

15. assistant (2026-05-31T04:25:43.028Z)

I have all the evidence needed. Let me finalize the verification.

****Final assessment against the checklist:****

1. ****Supported by the quote?**** Yes. The supporting quote (the "Complementary Material" definition) is verbatim accurate, and the MODEL_LICENSE explicitly adds "Complementary Material is not licensed under this License." The code/weights separation is real and documented.
2. ****Contradicting evidence?**** The one wrinkle: the claim characterizes "GPL/commercial dual-license" as merely what "older descriptions cite" (framing it as an outdated misread). The evidence shows GPL was *genuinely* Surya's prior code license (older README: "our code is GPL," \$2M threshold; even earlier weights were CC-BY-NC-SA-4.0). So the licensing genuinely **changed** ? it wasn't a misdescription. This is a minor imprecision in the claim's framing, not a refutation of its substance. The current state (Apache code + OpenRAIL-M weights) is exactly as claimed.
3. ****Source quality?**** Excellent ? primary sources (the actual LICENSE and MODEL_LICENSE files in the repo, plus README), corroborated by search.
4. ****Outdated?**** No. Verified against current `master` branch (2025-2026). The claim is explicitly **about** the current state vs. older descriptions, and gets the current state right.
5. ****Marketing/cherry-picked?**** No. Primary license documents.

****Verdict:**** The claim's three load-bearing assertions are all confirmed: (a) code LICENSE is Apache-2.0; (b) MODEL_LICENSE is OpenRAIL-M and explicitly excludes source/scripts; (c) the binding constraint for commercial-license-safety is the weights (OpenRAIL-M, with revenue/funding/AUP restrictions), so checking the repo license alone would wrongly conclude "fully permissive." This is precisely the correct conclusion for muneem's purpose, and it directly supports flagging Surya's weights as commercially-restricted. The only imprecision (treating the historical GPL as a "misdescription" rather than a genuine prior license) is peripheral and does not undermine the claim. The claim survives.

16. user (2026-05-31T04:25:53.586Z)

Structured output provided successfully

17. assistant (2026-05-31T04:25:59.758Z)

Verification complete. The claim is NOT refuted (confidence: high).

All three core assertions hold against primary sources on the current `datalab-to/surya` master branch:

- Code `LICENSE` = Apache-2.0 (verified verbatim header)
- `MODEL\_LICENSE` = "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)" and explicitly states "Complementary Material is not licensed under this License" ? confirming the code/scripts exclusion the supporting quote describes
- The weights are the binding commercial-safety constraint (OpenRAIL-M with >\$5M revenue/funding thresholds, competitor restriction, and use-based restrictions ? not freely commercial like Apache/MIT), so checking the repo license alone would wrongly conclude "fully permissive"

One minor imprecision noted but not refuting: GPL was a *genuine* prior code license for Surya (older README: "our code is GPL"), and an even earlier version used CC-BY-NC-SA-4.0 weights ? so the license materially changed over time rather than older sources being simply wrong. This is peripheral framing; the claim's substantive conclusions about the current state and the binding weights constraint are all correct and directly support flagging Surya's weights as commercially-restricted for muneem.